

MEZBAUR RAHMAN

✉ mezbaur00797@gmail.com 🏠 [Website](#) 🐙 [Github](#) 🔗 [Linkedin](#)

RESEARCH INTERESTS

Large Language Models | Vision Language Models | Learning from Noisy Labels | Semi Supervised Learning

EDUCATION

University of Illinois Chicago Ph.D. in Computer Science and Engineering Current CGPA: 4.00 out of 4.00	<i>Chicago, United States</i> Aug 2023 - Present
Islamic University of Technology M.Sc in Computer Science and Engineering CGPA: 3.96 out of 4.00	<i>Gazipur, Bangladesh</i> Jan 2020 - June 2023
Islamic University of Technology B.Sc in Computer Science and Engineering CGPA: 3.86 out of 4.00 (4 th in class)	<i>Gazipur, Bangladesh</i> Jan 2016 - Nov 2019

PROFESSIONAL EXPERIENCE

University of Illinois Chicago Graduate Research Assistant — advised by Prof. Cornelia Caragea Research in Semi-Supervised Learning & Learning from Noisy Labels with LLM Guidance	<i>Chicago, United States</i> Aug 2023 – Present
University of Illinois Chicago Graduate Teaching Assistant CS 521: Statistical NLP, CS 401: Computer Algorithms I, CS 251: Data Structure	<i>Chicago, United States</i> Spring 2024, Spring 2025
Islamic University of Technology Lecturer, Department of Computer Science and Engineering	<i>Gazipur, Bangladesh</i> Jan 2020 - July 2023
Samsung R&D Institute Bangladesh Software Engineering Intern	<i>Dhaka, Bangladesh</i> Nov 2018 - Jan 2019

PUBLICATIONS

- LLM-Guided Co-Training for Text Classification** 2025
EMNLP 2025 (Accepted at Main)
- We introduce a novel weighted co-training framework guided by Large Language Models (LLMs), where two encoder-only networks iteratively train each other using dynamically assigned sample weights based on confidence in LLM-generated pseudo-labels. Our method achieves state-of-the-art performance on 4 out of 5 benchmark datasets and ranks first among 14 SSL methods via Friedman test, demonstrating LLMs as effective knowledge amplifiers in semi-supervised learning.
arXiv Link: <https://arxiv.org/pdf/2509.16516>
Keyword: Semi-Supervised Learning, Large Language Models, Co-Training, Text Classification
- Multihop Factual Claim Verification Using Natural Language Prompts** 2023
Canadian AI 2023
- This research aims to establish a strategy for verifying claims based on evidence sentences by employing prompt fine-tuning of state-of-the-art pre-trained language models. This study's objectives also include developing suitable language prompts for this task. This research also investigates how using multiple sentences as evidence increases the difficulty of validating claims.
Online Link: <https://caiac.pubpub.org/pub/ex7vouwq/release/1>
Keyword: Natural Language Processing, Pre-trained Language Models, Prompt Fine Tuning

LREC 2026 (Submitted)

- This work investigates how semantic meanings of labels can drift across languages and cultures during translation, leading to inconsistencies in multilingual NLP benchmarks. We present a large-scale analysis supported by human annotation and propose alignment-aware evaluation protocols to mitigate cross-cultural semantic drift.
arXiv Link: <https://www.arxiv.org/pdf/2510.25967>
Keyword: Cross-Cultural Semantics, Translation, Multilingual NLP, Label Drift

Explainable Artificial Intelligence Model for Stroke Prediction Using EEG Signal 2022

2022

Sensors Journal

- This study follows an explainable approach to predicting stroke patients based on their biomarker data collected from EEG signals via various machine learning models.
Online Link: <https://www.mdpi.com/2008048>
Keyword: Electroencephalography, Stroke, Neuroscience, Machine-learning, Explainable AI

BanglaRQA: A Benchmark Dataset for Under-resourced Bangla Language Reading Comprehension-based Question Answering with Diverse Question-Answer Types

2022

Findings of EMNLP 2022

- This paper introduces a novel reading comprehension-based question-answer dataset containing 3000 Bangla Wikipedia context passages and 14889 question-answer pairings. The experiments in this work also improve the performance of a pre-trained transformer model, as evidenced by higher EM(exact match) and F1 scores when compared to previous work on other comparable Bangla datasets.
Online Link: <https://aclanthology.org/2022.findings-emnlp.186/>
Keywords: Natural Language Processing, Question-Answering, Transformer Models, BanglaT5

SKILLS

Programming Languages	Python, C/C++, Java, Shell Scripting
Databases	MySQL, Oracle SQL
Libraries & Frameworks	PyTorch, Kubernetes, Numpy, Matplotlib, Hugging Face, vLLM, Docker
Other	Version Control - Git, Latex/Overleaf

ONLINE CERTIFICATION

-
- **Neural Networks and Deep Learning** by DeepLearning.AI on Coursera
 - **Natural Language Processing with Probabilistic Models** by DeepLearning.AI on Coursera
 - **Natural Language Processing with Classification and Vector Spaces** by DeepLearning.AI on Coursera

SIGNIFICANT PROJECTS

-
- **Target Invariant Stance Detection Using Large Language Models:** In this project, we explore the performance of Large Language Models (LLMs) in stance detection, comparing the zero-shot capabilities of 7B models to fine-tuned smaller models. Our findings reveal trade-offs between model size, fine-tuning, and contextual understanding in NLP tasks. ([Github Link](#))
 - **LLM-Guided Node Classification in Semi-Supervised Settings:** This project integrates an LLM-based sentence encoder with Text-Attributed Graphs to enhance node features and employs LLM-guided pseudo labels to initiate a semi-supervised learning approach. Our results demonstrate that the LLM-guided approach excels with a larger proportion of unlabeled nodes, while the sentence encoder-based node features consistently improve overall performance. ([Github Link](#))